Solutions
QA : Algebra - 06

1. 3 Only the last 3 terms have numbers greater than or equal to 1 inside the [ ] sign. The last three terms are:

$$\left[\frac{1}{4}+\frac{38}{50}\right]+\left[\frac{1}{4}+\frac{39}{50}\right]+\left[\frac{1}{4}+\frac{40}{50}\right]$$

Each of these terms are equal to 1.
All previous terms have numbers that lie between 0 and 1 and therefore, are equal to zero.
Hence, the sum of the given terms is 3.

2. 1 Here, if x is an integer, then
[x] = x
So, x = x + 1, which is not possible.
If x is in the form of Integer (I) + Fraction (F)
then [x] = I
So, I = I + F + 1 = (I + 1) + F, which is again not possible.
Hence, no solution.

3. 2 Let x = [x] + {x} = I + f, where [x] = I denotes the integral part of x and {x} = f denotes the fractional part of x.

$$2[x]^2 = 5x + 2$$

$$\Rightarrow 2I^2 = 5I + 5f + 2$$

$$\Rightarrow f = \frac{2I^2 - 5I - 2}{5}$$

$$\Rightarrow 0 \le \frac{2I^2 - 5I - 2}{5} < 1$$

Solving the above inequality the only positive integral value of I that satisfies the equation is I = 3, and the corresponding value of f for this value of I is 0.2.
So, x = 3 + 0.2 = 3.2.
Hence, there is only one integral value of x that satisfies the given equation.

4. 2 As we have to maximise one integer.
So, we will assume the numbers are 1, 2, 3, 4....,14, a (where a is maximum)
$\Rightarrow$ 1 + 2 + 3 + ...+ 14 + a = 15 × 13
$\Rightarrow$ 7 × 15 + a = 15 × 13
$\Rightarrow$ a = 15(13 – 7) = 15 × 6 = 90.

5. 4 To get the minimum value, we will take first 9 natural numbers and 10th number (M) will be maximum such that $\frac{S}{M}$ could be maximized.
S = 1 + 2 + 3 + ... + 9 + M
S = 49 + M
Here, if we take M = 100

$$\Rightarrow \frac{S}{M} = \frac{149}{100} = 1.49$$

if we further increase the value of M,

$$\frac{S}{M} \to 1$$

So, minimum value of $\frac{S}{M} \approx 1$.

6. a + b + c = 10
We know that
AM(a, b, c) $\ge$ GM(a, b, c)

$$\Rightarrow \frac{a+b+c}{3} \ge (abc)^{1/3}$$

$$\Rightarrow \frac{10}{3} \ge (abc)^{1/3} \Rightarrow \frac{1000}{27} \ge abc$$

Hence, the maximum possible value of abc is $\frac{1000}{27}$.

7. a + b + c = 10
(a – 1) + (b – 2) + (c – 5) = 2
We know that
AM(a – 1, b – 2, c – 5) $\ge$ GM(a – 1, b – 2, c – 5)

$$\Rightarrow \frac{(a-1)+(b-2)+(c-5)}{3} \ge \{(a-1)\times(b-2)\times(c-5)\}^{1/3}$$

$$\Rightarrow \frac{2}{3} \ge \{(a-1)\times(b-2)\times(c-5)\}^{1/3}$$

$$\Rightarrow \frac{8}{27} \ge \{(a-1)\times(b-2)\times(c-5)\}$$

Hence, the maximum possible value of (a – 1)(b – 2)(c – 5) is $\frac{8}{27}$.

8. Since, a + b + c = 6
and AM $\ge$ GM

$$\text{So, } \frac{\frac{a}{3}+\frac{a}{3}+\frac{a}{3}+\frac{b}{2}+\frac{b}{2}+c}{6} \ge \left(\frac{a^3b^2c}{3^3 2^2}\right)^{\frac{1}{5}}$$

$$\Rightarrow \frac{6}{6} \ge \left(\frac{a^3b^2c}{3^3 2^2}\right)^{\frac{1}{5}} \Rightarrow 3^3 2^2 \ge a^3b^2c$$

So, maximum value of given expression is 108.

9. 4 x + y + z + w = 29
(x – 1) + (y + 3) + (z – 1) + (w – 2) = 28
Applying A.M. $\ge$ G.M.,

$$\frac{(x-1)+(y+3)+(z-1)+(w-2)}{4} \ge ((x-1)(y+3)(z-1)(w-2))^{1/4}$$

Maximum value of (x – 1) (y + 3) (z – 1) (w – 2)

$$= \left(\frac{28}{4}\right)^4 = 2401.$$

10. 2 The possible sets for distinct integers (x, y, z) are (1, 2, 8); (1, 3, 7); (1, 4, 6); (2, 3, 6); (2, 4, 5).
So, the maximum value of (xyz + xy + yz + zx) occurs for x = 2, y = 4 and z = 5.
Hence, the maximum value of given

11. 4 $\frac{p}{q}, \frac{q}{r}, \frac{r}{s}$ and $\frac{s}{p}$ are all positive numbers and for positive numbers.

A.M. $\geq$ G.M.

So, $\frac{\frac{p}{q}+\frac{q}{r}+\frac{r}{s}+\frac{s}{p}}{4} \geq \left(\frac{p}{q}\cdot\frac{q}{r}\cdot\frac{r}{s}\cdot\frac{s}{p}\right)^{\frac{1}{4}}$

or, $\frac{p}{q}+\frac{q}{r}+\frac{r}{s}+\frac{s}{p} \geq 4$

12. 3 Since a, b, c and d are positive real numbers.

So, $\frac{a+b+c+d}{4} \geq (abcd)^{\frac{1}{4}}$

$\Rightarrow$ a + b + c + d $\geq$ 4

Again,

$$\frac{(a+1)+(b+1)+(c+1)+(d+1)}{4} \geq [(a+1)(b+1)(c+1)(d+1)]^{1/4}$$

$$\Rightarrow \frac{a+b+c+d}{4}+1 \geq [(a+1)(b+1)(c+1)(d+1)]^{1/4}$$

Since, we have to minimise RHS, so LHS must be minimum for minimum value of LHS

a + b + c + d = 4

$$\Rightarrow \frac{4}{4}+1=\left[(a+1)(b+1)(c+1)(d+1)\right]^{1/4}$$

$$16=\left[(a+1)(b+1)(c+1)(d+1)\right]_{min.}$$

Alternative method:

For minimum value a = b = c = d = 1

So, (1 + a) (1 + b) (1 + c) (1 + d) = 2 × 2 × 2 × 2 = 16.

13. 3 Since x + y = 1

So, for the minimum value of given expression x = y = 1/2

We put this value on the given expression.

So, $\left(\frac{1}{2}+2\right)^2+\left(\frac{1}{2}+2\right)^2=(2.5)^2+(2.5)^2=12.5$

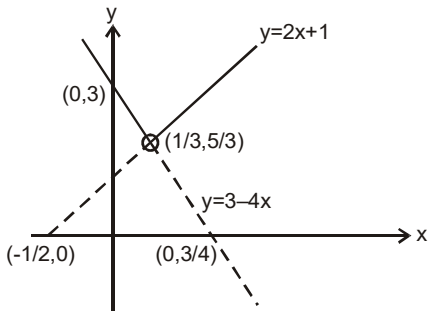
14. 4 f(x) = max (2x + 1, 3 – 4x)

So, the two equations are y = 2x + 1 and y = 3 – 4x

Their point of intersection would be

2x + 1 = 3 – 4x

$\Rightarrow$ 6x = 2 $\Rightarrow$ $x=\frac{1}{3}$



So when $x \leq \frac{1}{3}$, then $f(x)_{max}=3-4x$

and when $x \geq \frac{1}{3}$, then $f(x)_{max}=2x+1$

Hence, the minimum of this will be at x = $\frac{1}{3}$

i.e. $y=\frac{5}{3}$.

15. 2 It is given that abc = 4.

We know, (a + b + c) will be minimum when a, b, c are constants and a = b = c.

As a, b, c are in arithmetic progression,

$\Rightarrow$ a + b + c = 3b

This will be minimum when a = b = c, $b^3 = 4$.

$\Rightarrow b=4^{1/3}=2^{2/3}$

16. 3 S_1, S_2, S_3 and S_4 are (a + b + c), (b + c + d), (a + d + c) and (a + b + d). As all the sums are positive, using the AM $\geq$ GM property, we get,

$\frac{a+b+c}{3} \geq (abc)^{1/3}$ … (i)

$\frac{b+c+d}{3} \geq (bcd)^{1/3}$ … (ii)

$\frac{c+d+a}{3} \geq (cda)^{1/3}$ … (iii)

$\frac{a+d+b}{3} \geq (adb)^{1/3}$ … (iv)

Multiplying (i), (ii), (iii) and (iv), we get

(a + b + c)(b + c + d)(c + d + a)(a + d + b) $\geq 3^4$ (abcd)

$S_1 \times S_2 \times S_3 \times S_4 \geq 405$.

17. 2 Number of samosas = 200 + 20n,

n is a natural number.

Price per samosa = Rs. (2 – 0.1n)

Revenue = (200 + 20n)(2 – 0.1n) = 400 + 20n – $2n^2$

For maxima, n = 5

$\Rightarrow$ Maximum revenue will be at (200 + 20 × 5)

= 300 samosas

18. 4 $\sqrt{x}-\sqrt{1-x}=\frac{1}{5}$

$\sqrt{x}+\sqrt{1-x}=a$, say

On squaring and adding, we get

$2(x + 1 - x) = \frac{1}{25} + a^2$

$\Rightarrow a^2=2-\frac{1}{25}=\frac{49}{25}$

$\Rightarrow a=\frac{7}{5}$ (as $a > 0$)

19. 3 Given pqr = 1

$\Rightarrow \quad pq=\frac{1}{r}$ and $\frac{1}{p}=qr$

$$\frac{1}{1+p+q^{-1}}+\frac{1}{1+q+r^{-1}}+\frac{1}{1+r+p^{-1}}$$

$$=\frac{q}{1+q+pq}+\frac{r}{1+qr+r}+\frac{1}{1+r+qr}$$

$$=\frac{qr}{1+qr+r}+\frac{r}{1+qr+r}+\frac{1}{1+r+qr}=\frac{1+r+qr}{1+r+qr}=1.$$

Alternative method:

Putting x = y = z = 1, we get

$$\frac{1}{1+p+q^{-1}}+\frac{1}{1+q+r^{-1}}+\frac{1}{1+r+p^{-1}}$$

$$=\frac{1}{1+1+1}+\frac{1}{1+1+1}+\frac{1}{1+1+1}=\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=1.$$

20. 4 $\frac{1}{m}+\frac{4}{n}=\frac{1}{12}, n<60$

$\Rightarrow \frac{1}{m}=\frac{1}{12}-\frac{4}{n}=\frac{n-48}{12n}$

$\Rightarrow m=\frac{12n}{n-48}$

Positive integral values of m for odd integral values of n are for n = 49, 51 and 57.

Therefore, there are 3 integral pairs of values of m and n that satisfy the given equation.

21. 3 Given equation is x + y = xy

$\Rightarrow$ xy – x – y + 1 = 1

$\Rightarrow$ (x – 1)(y – 1) = 1

x – 1 = 1 & y – 1 = 1 or x – 1 = –1 & y – 1 = –1

Clearly (0, 0) and (2, 2) are the only pairs that will satisfy the equation.

22. 2 Minimum of sum of three squares should theoretically be zero if a = b = c = d.

But for that a = b = c = d = 7.5 (not possible as a, b, c, d are integers).

Thus Assume a = 8.

Let us assume b = 8, c = 9, then d = 7.

Thus, we get

$(a - b)^2 = 0$, $(a - c)^2 = 1$ $(a - d)^2 = 1$

$\therefore$ Required Sum = 2.

23. 1 Given ab = 432, bc = 96 and c < 9, where a, b and c are positive integers.

The value of c should be 8, 6, 4 or 2.

At c = 8, b = 12 and a = 36

At c = 6, b = 16 and a = 27

At c = 4, b = 24 and a = 18

At c = 2, b = 48 and a = 9

Hence, minimum value of a + b + c = 18 + 24 + 4 = 46.

24. 2704

As x + y = 102 (given)

Now, 2601 = 51 × 51

So, $2601\left(1+\frac{1}{x}\right)\left(1+\frac{1}{y}\right)$ can be written as,

$$=51\times 51\left(1+\frac{1}{x}\right)\left(1+\frac{1}{y}\right)$$

$$=\left(51+\frac{51}{x}\right)\left(51+\frac{51}{y}\right)$$

We have to find the minimum possible value of the given expression, therefore put x = y = 51

$$\therefore\left(51+\frac{51}{51}\right)\left(51+\frac{51}{51}\right)=(52)(52)=2704.$$

25. 34 xy + yz = 19

$\Rightarrow$ y(x + z) = 19 … (i)

19 is prime $\Rightarrow$ Either y = 19

and (x + z) = 1

or (x + z) = 19 and y = 1

yz + xz = 51 … (ii)

z(y + x) = 3 × 17

Either y = 3 and (y + x) = 17

or z = 17 and (y + x) = 3

Case 1: z = 17 (x, y) = (1, 2)

$\Rightarrow$ xyz = 34

Case 2: z = 3 (x, y) = (16, 1)

$\Rightarrow$ xyz = 48

z = 3 (x, y) = (15, 2)

$\Rightarrow$ xyz = 90

On checking for other possible values of x, y and z we can observe that the minimum possible value of xyz = 34.

26. 3 $c = \frac{16x}{y} + \frac{49y}{x}$

Let k = x/y, since x and y are real, k is also real.

So $c = 16k + \frac{49}{k}$

$\Rightarrow 16k^2 - ck + 49 = 0$

For k to be real, it has to be rational first.
This means, $c^2 > (2 \times 4 \times 7)^2$
or, $c^2 > 56^2$
or, $|c| > 56$
Hence, c cannot take the value –50.

27. 4 $\frac{f(x)}{g(x)} = \frac{\left(\frac{x^2+1}{x^2-1}\right)}{\left(\frac{x+1}{x-1}\right)} = \frac{x^2+1}{(x+1)^2} > 0.$

Let y = $\frac{x^2+1}{(x+1)^2}$

$\Rightarrow x^2(y - 1) + 2yx + (y - 1) = 0$ which is a quadratic equation in x.
Discriminant should be greater than 0.
$4y^2 - 4(y - 1)^2 \geq 0$

$\Rightarrow y \geq \frac{1}{2}$

And x = 1, $\frac{f(x)}{g(x)} = \frac{1}{3}$

Hence, either the value should be greater than 1/2 or should be equal to 1/3.

28. 32 f(x) = min $\{2x^2, 52 - 5x\}$

The maximum possible value of f(x) is obtained when $52 - 5x = 2x^2$.

Solving further we get,

(x – 4) (2x + 13) = 0

x = 4 and $x = -\frac{13}{2}$

$x = -\frac{13}{2}$ will be rejected since x is a positive real number.

At x = 4, f(x) = 32.

For questions 29 and 30:
To maximise Shabnam's return we need to evaluate all the given options in the question number 7. Assume Shabnam had one rupee to invest. Let the return be denoted by 'r'.
Consider the option (30% in option A, 32% in option B and 38% in option C): If the stock market rises, then
r = 0.1 × 0.3 + 5 × 0.32 – 2.5 × 0.38 = 0.653
If the stock market falls, then
r = 0.1 × 0.3 – 3 × 0.32 + 2 × 0.38 = – 0.197
Consider option (100% in option A): This will give a return of 0.1%.
Consider option (36% in option B and 64% in option C):
If the stock market rises, then
r = 5 × 0.36 – 2.5 × 0.64 = 0.2
If the stock market falls, then
r = – 3 × 0.36 + 2 × 0.64 = 0.2
Consider option (64% in option B and 36% in option C):
If the stock market rises, then
r = 5 × 0.64 – 2.5 × 0.36 = 2.1
If the stock market falls, then
r = – 3 × 0.64 + 2 × 0.36 = –1.2
Consider option (1/3 in each of the 3 options): If the stock market rises, then
r = 0.1 × 0.33 + 5 × 0.33 – 2.5 × 0.33 = 0.858
If the stock market falls, then
r = 0.1 × 0.33 – 3 × 0.33 + 2 × 0.33 = –0.297
We can see that only in option **(36% in option B and 64% in option C)**, Shabnam gets an assured return of 0.2% irrespective of the behaviour of the stock market. So right option for questions number 13 is **(0.20%)** and question number 14 is **(36% in option B and 64% in option C)**.

29. 3

30. 2